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Introduction

- في هذا المشروع تم تصميم شبكة تتكون من ثلاثة مباني، كل مبنى يحتوي على ثلاثة طوابق، كل طابق متصل بـ Access Switch خاص به، وجميع الـ Access Switches متصلة بـ Multilayer Switch في نفس المبنى، يقوم الـ Multilayer Switch في كل مبنى بعملية Routing داخلي بين الـ VLANs والشبكات الفرعية (Subnets) الخاصة بالطوابق الثلاثة، مما يتيح للأجهزة داخل المبنى التواصل مع بعضها البعض.
- أما التواصل بين المباني المختلفة، فيتم من خلال Edge Router، حيث أن كل Multilayer Switch متصل بالـ Edge Router، وهو بدوره مسؤول عن:
 - تمرير الترافيك بين المباني المختلفة (Inter-Building Communication).
 - توصيل الشبكة الداخلية بالـ ISP (Internet Service Provider).
 - تطبيق NAT لترجمة العناوين الخاصة الداخلية إلى عنوان عام (Public IP).
- بهذا التصميم:
 - يتم تحقيق الاتصال الداخلي بين الطوابق عبر الـ Multilayer Switch.
 - يتم تحقيق الاتصال بين المباني والخروج إلى الإنترنت عبر الـ Edge Router.
 - تتوافر خدمات الشبكة مثل DHCP, DNS, FTP, Web, Email، بالإضافة إلى تطبيق سياسات الحماية (Port Security, ACLs).

◆ Build 1

→ Floor 1:

- Number Of Hosts :** 16 جهاز PC.
- VLAN:** VLAN 10 (Subnet 10.0.0.0/27).
- Gateway (Virtual IP):** 10.0.0.1 (Multilayer Switch – Active/Standby HSRP مُعدّ على).
- Broadcast:** 10.0.0.31.
- DHCP:** Multilayer Switch (Active) موزع من الـ.
- Usable IP Addresses:** 30
- Used IPs:** 16 PCs + 3 reserved addresses (Active Gateway, Standby Gateway, Virtual IP) = 19
- Available IPs (not used):** 11
- PC0:** is assigned a static IP = 10.0.0.4.
 - Reason: This device will be the target for an Access Control List (ACL) test case.
- Access Switch**
 - Model:** Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports).
 - Access Ports:**
 - Interfaces Fa0/3 – Fa0/18 configured as Access Mode, assigned to VLAN 10 (for connecting PCs).
 - Trunk Ports:**
 - Interfaces Fa0/1 – Fa0/2 configured as Trunk Mode, used as uplinks to the Multilayer Switch.
 - Unused Ports:**
 - Fa0/19 – Fa0/24, Gi0/1 – Gi0/2 → currently unused and available for future expansion.
 - Interface Fa0/5 reserved for Port Security (Sticky MAC):** ``

```
interface fa0/5 -
switchport mode access -
switchport port-security -
switchport port-security maximum 1 -
```

** → Floor 2:

- **Number Of Hosts :** 10 جهاز PC.
- **VLAN:** VLAN 20 (Subnet 10.0.0.96/28).
- **Gateway (Virtual IP):** 10.0.0.97 (مُعدّ على Multilayer Switch – Active/Standby HSRP).
- **Broadcast:** 10.0.0.111.
- **DHCP:** موزع من ال Multilayer Switch (Active).
- **Usable IP Addresses:** 14
- **Used IPs:** 10 PCs + 3 reserved addresses (Active Gateway, Standby Gateway, Virtual IP) = 13
- **Available IPs (not used):** 1
- **Access Switch**
 - **Model:** Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports).
 - **Access Ports:**
 - Interfaces Fa0/3 – Fa0/12 configured as Access Mode, assigned to VLAN 20 (for connecting PCs).
 - **Trunk Ports:**
 - Interfaces Fa0/1 – Fa0/2 configured as Trunk Mode, used as uplinks to the Multilayer Switch.
 - **Unused Ports:**
 - Fa0/13 – Fa0/24, Gi0/1 – Gi0/2 → currently unused and available for future expansion.

** → Floor 3:

- **Number Of Hosts :** 12 جهاز PC.
- **VLAN:** VLAN 30 (Subnet 10.0.0.64/27).
- **Gateway (Virtual IP):** 10.0.0.65 (مُعدّ على Multilayer Switch – Active/Standby HSRP).
- **Broadcast:** 10.0.0.95.
- **DHCP:** موزع من ال Multilayer Switch (Active).
- **Usable IP Addresses:** 30
- **Used IPs:** 12 PCs + 3 reserved addresses (Active Gateway, Standby Gateway, Virtual IP) = 15
- **Available IPs (not used):** 15
- **Access Switch**
 - **Model:** Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports).
 - **Access Ports:**
 - Interfaces Fa0/3 – Fa0/14 configured as Access Mode, assigned to VLAN 30 (for connecting PCs).
 - **Trunk Ports:**
 - Interfaces Fa0/1 – Fa0/2 configured as Trunk Mode, used as uplinks to the Multilayer Switch.
 - **Unused Ports:**
 - Fa0/15 – Fa0/24, Gi0/1 – Gi0/2 → currently unused and available for future expansion.

→ Multilayer Switch 0 (Active) :

- **Model:** Cisco Catalyst 6500 (Multilayer Switch).
- **Role:** Active switch in HSRP (provides routing between floors of Building 1 and forwards traffic to the Edge Router).
- **VLANs and Interfaces**
 - **VLAN 10 (Floor 1):**
 - Interface IP: 10.0.0.2/27.
 - Virtual IP (HSRP): 10.0.0.1.
 - This virtual IP is the default gateway for all PCs in Floor 1.
 - **VLAN 20 (Floor 2):**
 - Interface IP: 10.0.0.98/28.

- Virtual IP (HSRP): 10.0.0.97.
 - Default gateway for Floor 2 PCs.
- **VLAN 30 (Floor 3):**
 - Interface IP: 10.0.0.66/27.
 - Virtual IP (HSRP): 10.0.0.65.
 - Default gateway for Floor 3 PCs.
- **Trunk Interfaces**
 - Gi1/0/1 – Gi1/0/3 configured as Trunk Ports, connecting the Multilayer Switch to the access switches of the three floors.
- **DHCP Configuration**
 - **Pool F1 (Floor 1):**
 - Network: 10.0.0.0/27
 - Default-router: 10.0.0.1
 - DNS: 10.0.0.180
 - Excluded addresses: 10.0.0.1 – 10.0.0.4
 - **Pool F2 (Floor 2):**
 - Network: 10.0.0.96/28
 - Default-router: 10.0.0.97
 - DNS: 10.0.0.180
 - Excluded addresses: 10.0.0.97 – 10.0.0.99
 - **Pool F3 (Floor 3):**
 - Network: 10.0.0.64/27
 - Default-router: 10.0.0.65
 - DNS: 10.0.0.180
 - Excluded addresses: 10.0.0.65 – 10.0.0.67
 - ⚠ **When configuring the DHCP pools, certain addresses were excluded to reserve them for static assignment.**
 - Example:
 - VLAN 10 (Floor 1): excluded-address 10.0.0.1 – 10.0.0.4
 - 10.0.0.1 → Virtual Gateway (HSRP)
 - 10.0.0.2 → Active MLS interface
 - 10.0.0.3 → Standby MLS interface
 - 10.0.0.4 → Reserved for PC0 (static) → used in ACL testing
- **Uplink to Edge Router**
 - Interface Gi1/0/4 is configured as a Routed Port:
 - IP address: 10.0.0.185/30
 - Connected directly to the Edge Router.
- **HSRP Configuration**
 - **Active Role:** This Multilayer Switch is configured as the Active HSRP device for VLANs 10, 20, and 30.
 - **Standby Role:** The second Multilayer Switch in Building 1 acts as the Standby HSRP device.
 - **Priority:** Set to 150 (higher than the default 100), ensuring that this switch always takes the Active role when available.
 - **Preempt Enabled:** With the preempt command configured, if this Active switch fails and later recovers, it will automatically reclaim the Active role.
- **Routing**
 - **Default Route:** Configured to forward unknown traffic to the Edge Router (Next-Hop: 10.0.0.186).
 - **OSPF:** Enabled for internal routing between VLANs and to share routes with the Edge Router, ensuring connectivity between different buildings.
- **Remote Access (Telnet)**
 - Telnet is enabled on the HSRP Virtual IP for VLANs.
 - When connecting via Telnet to the virtual IP (e.g., 10.0.0.1 for VLAN 10), the session will always be established with the Active Multilayer Switch.
 - This method allows quick verification of which switch is currently the Active HSRP device

→ Multilayer Switch 1 (Standby) :

- **Model:** Cisco Catalyst 6500 (Multilayer Switch).
- **Role:** Standby switch in HSRP (provides redundancy for the Active switch, takes over only if the Active fails).
- **VLANs and Interfaces**
 - **VLAN 10 (Floor 1):**
 - Interface IP: 10.0.0.3/27
 - Virtual IP (HSRP): 10.0.0.1
 - The virtual IP acts as the default gateway for all PCs in Floor 1.
 - **VLAN 20 (Floor 2):**
 - Interface IP: 10.0.0.99/28
 - Virtual IP (HSRP): 10.0.0.97
 - Default gateway for Floor 2 PCs.
 - **VLAN 30 (Floor 3):**
 - Interface IP: 10.0.0.67/27
 - Virtual IP (HSRP): 10.0.0.65
 - Default gateway for Floor 3 PCs.
- **Trunk Interfaces**
 - Gi1/0/1 – Gi1/0/3 configured as Trunk Ports, connecting the Multilayer Switch to the access switches of the three floors.
- **DHCP Configuration**
 - **Pool F1 (Floor 1):**
 - Network: 10.0.0.0/27
 - Default-router: 10.0.0.1
 - DNS: 10.0.0.180
 - Excluded addresses: 10.0.0.1 – 10.0.0.4 (Virtual IP + Active + Standby + Static PC0).
 - **Pool F2 (Floor 2):**
 - Network: 10.0.0.96/28
 - Default-router: 10.0.0.97
 - DNS: 10.0.0.180
 - Excluded addresses: 10.0.0.97 – 10.0.0.99 (Virtual IP + Active + Standby).
 - **Pool F3 (Floor 3):**
 - Network: 10.0.0.64/27
 - Default-router: 10.0.0.65
 - DNS: 10.0.0.180
 - Excluded addresses: 10.0.0.65 – 10.0.0.67 (Virtual IP + Active + Standby).
- **Uplink to Edge Router**
 - Interface Gi1/0/4 is configured as a Routed Port:
 - IP address: 10.0.0.189/30
 - Connected directly to the Edge Router.
- **HSRP Configuration**
 - **Standby Role:** This Multilayer Switch operates as the Standby HSRP device for VLANs 10, 20, and 30.
 - **No Priority/Preempt:** By default, it takes over only if the Active switch fails, and it does not reclaim Active role once the Active comes back online.
- **Routing**
 - **Default Route:** Configured to forward unknown traffic to the Edge Router (Next-Hop: 10.0.0.190).
 - **OSPF:** Enabled to advertise VLAN networks and learn routes from the Edge Router, ensuring inter-building connectivity.
- **Remote Access (Telnet)**
 - Telnet is configured but sessions through the Virtual IP are always directed to the Active switch.

→ Edge Router 0 :

- **Model:** PT-Router (Packet Tracer Default Router)

- **Role:** Connects the internal network of Building 1 to the ISP, applies NAT, and enforces ACL policies, It also acts as the interconnection point between different buildings, ensuring that devices in Building 1, Building 2, and Building 3 can communicate with each other through routing.
- **Interfaces and IPs**
 - FastEthernet 0/0 → 10.0.0.186/30 (connected to Active MLS)
 - FastEthernet 1/0 → 10.0.0.190/30 (connected to Standby MLS)
 - GigabitEthernet 9/0 → 20.0.0.1/30 (public-facing interface, connected to ISP)
 - Serial 2/0 → 10.0.0.209/30 (connected to router of build2)
- **Routing**
 - **Default Route:** 0.0.0.0/0 → Next-Hop 20.0.0.2 (ISP)
 - **OSPF:** Configured to advertise the routed links to the Multilayer Switches.
This allows all internal subnets of the three floors to reach the router through the MLS devices, and ensures communication between buildings as well as access to the ISP.
↪Advertised networks:
 - 10.0.0.184/30 (link to Active MLS)
 - 10.0.0.188/30 (link to Standby MLS)
 - 10.0.0.208/30 (link for backbone/ISP interconnection)
- **Access Control List (ACL)**
 - Access-list 110:
 - Deny TCP traffic from host 10.0.0.4 (PC0, static IP) to any FTP service (port 21).
 - Permit all other IP traffic.
 - Applied inbound on FastEthernet 0/0 interface.

```
access-list 110 deny tcp host 10.0.0.4 any eq ftp-
access-list 110 permit ip any any-
int f0/0-
ip access-group 110 in-
```

- **Network Address Translation (NAT)**
 - **Type:** NAT Overload (PAT) :
 - ACL 1 defines the inside subnets permitted for NAT:
 - 10.0.0.0/27 (Floor 1)
 - 10.0.0.96/28 (Floor 2)
 - 10.0.0.64/27 (Floor 3)
 - **Configuration:**
 - NAT Inside Interfaces: FastEthernet 0/0, FastEthernet 1/0
 - NAT Outside Interface: GigabitEthernet 9/0
 - Translation: **All inside networks share the public IP of Gi9/0 (20.0.0.1) using overload.**

```
access-list 1 permit 10.0.0.0 0.0.0.31-
access-list 1 permit 10.0.0.96 0.0.0.15-
access-list 1 permit 10.0.0.64 0.0.0.31-
ip nat inside source list 1 interface g9/0 overload-
int g9/0-
ip nat outside-
int f0/0-
ip nat inside-
int f1/0-
ip nat inside-
```

→ Floor 1:

- **Number Of Hosts** : 7 جهاز PC.
- **VLAN**: VLAN 40 (Subnet 10.0.0.144/28).
- **Gateway (Virtual IP)**: 10.0.0.145 (مُعدّ على Multilayer Switch – Active/Standby HSRP).
- **Broadcast**: 10.0.0.159.
- **DHCP**: موزع من ال Multilayer Switch (Active).
- **Usable IP Addresses**: 14
- **Used IPs**: 7 PCs + 3 reserved addresses (Active Gateway, Standby Gateway, Virtual IP) = 10
- **Available IPs (not used)**: 4
- **Access Switch**
 - **Model**: Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports).
 - **Access Ports**:
 - Interfaces Fa0/3 – Fa0/9 configured as Access Mode, assigned to VLAN 40 (for connecting PCs).
 - **Trunk Ports**:
 - Interfaces Fa0/1 – Fa0/2 configured as Trunk Mode, used as uplinks to the Multilayer Switch.
 - **Unused Ports**:
 - Fa0/10 – Fa0/24, Gi0/1 – Gi0/2 → currently unused and available for future expansion.

** → Floor 2:

- **Number Of Hosts** : 1 PC configured as a Server (multi-service)
- **VLAN**: VLAN 50 (Subnet 10.0.0.176/29).
- **Static ip** : 10.0.0.180
- **Gateway (Virtual IP)**: 10.0.0.177 (مُعدّ على Multilayer Switch – Active/Standby HSRP).
- **Broadcast**: 10.0.0.183.
- **Usable IP Addresses**: 6
- **Used IPs**: 1 PCs + 3 reserved addresses (Active Gateway, Standby Gateway, Virtual IP) = 4
- **Available IPs (not used)**: 2
- **Services enabled**:
 - **Web Server (HTTP)** – reachable by all clients.
 - **DNS Server** – domain name: www.mahmoud.com .
 - **Email Server (SMTP/POP3)** – configured with user accounts.
 - **FTP Server** – requires authentication for access.
 - **FTP and Email User Accounts**:
 - Username: ali , Password: 12345
 - Username: noor , Password: 12345
- **Access Switch**
 - **Model**: Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports).
 - **Access Ports**:
 - Interfaces Fa0/3 configured as Access Mode, assigned to VLAN 50 (for connecting Server).
 - **Trunk Ports**:
 - Interfaces Fa0/1 – Fa0/2 configured as Trunk Mode, used as uplinks to the Multilayer Switch.
 - **Unused Ports**:
 - Fa0/4 – Fa0/24, Gi0/1 – Gi0/2 → currently unused and available for future expansion.

** → Floor 3:

- **Number Of Hosts** : 5 جهاز PC.
- **VLAN**: VLAN 60 (Subnet 10.0.0.160/28).
- **Gateway (Virtual IP)**: 10.0.0.161 (مُعدّ على Multilayer Switch – Active/Standby HSRP).

- **Broadcast:** 10.0.0.175.
- **DHCP:** موزع من ال Multilayer Switch (Active).
- **Usable IP Addresses:** 14
- **Used IPs:** 5 PCs + **3 reserved addresses** (Active Gateway, Standby Gateway, Virtual IP) = 8
- **Available IPs (not used):** 6
- **Access Switch**
 - **Model:** Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports).
 - **Access Ports:**
 - Interfaces Fa0/3 – Fa0/7 configured as Access Mode, assigned to VLAN 60 (for connecting PCs).
 - **Trunk Ports:**
 - Interfaces Fa0/1 – Fa0/2 configured as Trunk Mode, used as uplinks to the Multilayer Switch.
 - **Unused Ports:**
 - Fa0/8 – Fa0/24, Gi0/1 – Gi0/2 → currently unused and available for future expansion.

→ Multilayer Switch 2 (Active) :

- **Model:** Cisco Catalyst 6500 (Multilayer Switch).
- **Role:** Active switch in HSRP (provides routing between floors of Building 1 and forwards traffic to the Edge Router).
- **VLANs and Interfaces**
 - **VLAN 40 (Floor 1):**
 - Interface IP: 10.0.0.146/28.
 - Virtual IP (HSRP): 10.0.0.145.
 - This virtual IP is the default gateway for all PCs in Floor 1.
 - **VLAN 50 (Floor 2):**
 - Interface IP: 10.0.0.178/29.
 - Virtual IP (HSRP): 10.0.0.177.
 - Default gateway for Floor 2 PCs.
 - **VLAN 60 (Floor 3):**
 - Interface IP: 10.0.0.162/28.
 - Virtual IP (HSRP): 10.0.0.161.
 - Default gateway for Floor 3 PCs.
- **Trunk Interfaces**
 - Gi1/0/1 – Gi1/0/3 configured as Trunk Ports, connecting the Multilayer Switch to the access switches of the three floors.
- **DHCP Configuration**
 - **Pool F4 (Floor 1):**
 - Network: 10.0.0.144/28
 - Default-router: 10.0.0.145
 - DNS: 10.0.0.180
 - Excluded addresses: 10.0.0.145 – 10.0.0.147
 - **Pool F6 (Floor 3):**
 - Network: 10.0.0.160/28
 - Default-router: 10.0.0.161
 - DNS: 10.0.0.180
 - Excluded addresses: 10.0.0.161 – 10.0.0.163
- **Uplink to Edge Router**
 - Interface Gi1/0/4 is configured as a Routed Port:
 - IP address: 10.0.0.193/30.
 - Connected directly to the Edge Router.
- **HSRP Configuration**
 - **Active Role:** This Multilayer Switch is configured as the Active HSRP device for VLANs 40, 50, and 60.
 - **Standby Role:** The second Multilayer Switch in Building 2 acts as the Standby HSRP device.

- **Priority:** Set to 150 (higher than the default 100), ensuring that this switch always takes the Active role when available.
- **Preempt Enabled:** With the preempt command configured, if this Active switch fails and later recovers, it will automatically reclaim the Active role.
- **Routing**
 - **Default Route:** Configured to forward unknown traffic to the Edge Router (Next-Hop: 10.0.0.194).
 - **OSPF:** Enabled for internal routing between VLANs and to share routes with the Edge Router, ensuring connectivity between different buildings.

→ Multilayer Switch 3 (Standby) :

- **Model:** Cisco Catalyst 6500 (Multilayer Switch).
- **Role:** Standby switch in HSRP (provides redundancy for the Active switch, takes over only if the Active fails).
- **VLANs and Interfaces**
 - **VLAN 40 (Floor 1):**
 - Interface IP: 10.0.0.147/28.
 - Virtual IP (HSRP): 10.0.0.145.
 - This virtual IP is the default gateway for all PCs in Floor 1.
 - **VLAN 50 (Floor 2):**
 - Interface IP: 10.0.0.179/29.
 - Virtual IP (HSRP): 10.0.0.177.
 - Default gateway for Floor 2 PCs.
 - **VLAN 60 (Floor 3):**
 - Interface IP: 10.0.0.163/28.
 - Virtual IP (HSRP): 10.0.0.161.
 - Default gateway for Floor 3 PCs.
- **Trunk Interfaces**
 - Gi1/0/1 – Gi1/0/3 configured as Trunk Ports, connecting the Multilayer Switch to the access switches of the three floors.
- **DHCP Configuration**
 - **Pool F4 (Floor 1):**
 - Network: 10.0.0.144/28
 - Default-router: 10.0.0.145
 - DNS: 10.0.0.180
 - Excluded addresses: 10.0.0.145 – 10.0.0.147
 - **Pool F6 (Floor 3):**
 - Network: 10.0.0.160/28
 - Default-router: 10.0.0.161
 - DNS: 10.0.0.180
 - Excluded addresses: 10.0.0.161 – 10.0.0.163
- **Uplink to Edge Router**
 - Interface Gi1/0/4 is configured as a Routed Port:
 - IP address: 10.0.0.197/30
 - Connected directly to the Edge Router.
- **HSRP Configuration**
 - **Standby Role:** This Multilayer Switch operates as the Standby HSRP device for VLANs 40, 50, and 60.
 - **No Priority/Preempt:** By default, it takes over only if the Active switch fails, and it does not reclaim Active role once the Active comes back online.
- **Routing**
 - **Default Route:** Configured to forward unknown traffic to the Edge Router (Next-Hop: 10.0.0.198).
 - **OSPF:** Enabled to advertise VLAN networks and learn routes from the Edge Router, ensuring inter-building connectivity.

→ Edge Router 1 :

- **Model:** PT-Router (Packet Tracer Default Router)

- **Role:** Connects the internal network of Building 2 to the ISP, applies NAT, and It also acts as the interconnection point between different buildings, ensuring that devices in Building 1, Building 2, and Building 3 can communicate with each other through routing.
- **Interfaces and IPs**
 - FastEthernet 0/0 → 10.0.0.194/30 (connected to Active MLS)
 - FastEthernet 1/0 → 10.0.0.198/30 (connected to Standby MLS)
 - GigabitEthernet 9/0 → 20.0.0.5/30 (public-facing interface, connected to ISP)
 - Serial 2/0 → 10.0.0.210/30 (connected to router of build1)
 - Serial 3/0 → 10.0.0.213/30 (connected to router of build3)
- **Routing**
 - **Default Route:** 0.0.0.0/0 → Next-Hop 20.0.0.6 (ISP)
 - **OSPF:** Configured to advertise the routed links to the Multilayer Switches.
This allows all internal subnets of the three floors to reach the router through the MLS devices, and ensures communication between buildings as well as access to the ISP.
↪Advertised networks:
 - 10.0.0.192/30 (link to Active MLS)
 - 10.0.0.196/30 (link to Standby MLS)
 - 10.0.0.208/30 (link for build2)
 - 10.0.0.212/30 (link for build3)
- **Network Address Translation (NAT)**
 - **Type:** NAT Overload (PAT) :
 - ACL 1 defines the inside subnets permitted for NAT:
 - 10.0.0.144/28 (Floor 1)
 - 10.0.0.176/29 (Floor 2)
 - 10.0.0.160/28 (Floor 3)
 - **Configuration:**
 - NAT Inside Interfaces: FastEthernet 0/0, FastEthernet 1/0
 - NAT Outside Interface: GigabitEthernet 9/0
 - Translation: **All inside networks share the public IP of Gi9/0 (20.0.0.5) using overload.**

```
access-list 1 permit 10.0.0.144 0.0.0.15 -
access-list 1 permit 10.0.0.176 0.0.0.7 -
access-list 1 permit 10.0.0.160 0.0.0.15 -
ip nat inside source list 1 interface g9/0 overload -
int g9/0 -
ip nat outside -
int f0/0 -
ip nat inside -
int f1/0 -
ip nat inside -
```

◆ Build 3

➔ Floor 1:

- **Number Of Hosts :** 8 جهاز PC.
- **VLAN:** VLAN 70 (Subnet 10.0.0.128/28).
- **Gateway (Virtual IP):** 10.0.0.129 (مُعدّ على Multilayer Switch – Active/Standby HSRP).
- **Broadcast:** 10.0.0.143.
- **DHCP:** موزع من ال Multilayer Switch (Active).
- **Usable IP Addresses:** 14

- **Used IPs:** 8 PCs + 3 reserved addresses (Active Gateway, Standby Gateway, Virtual IP) = 11
- **Available IPs (not used):** 3
- **Access Switch**
 - **Model:** Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports).
 - **Access Ports:**
 - Interfaces Fa0/3 – Fa0/10 configured as Access Mode, assigned to VLAN 70 (for connecting PCs).
 - **Trunk Ports:**
 - Interfaces Fa0/1 – Fa0/2 configured as Trunk Mode, used as uplinks to the Multilayer Switch.
 - **Unused Ports:**
 - Fa0/11 – Fa0/24, Gi0/1 – Gi0/2 → currently unused and available for future expansion.

**** → Floor 2:**

- **Number Of Hosts :** 14 جهاز PC.
- **VLAN:** VLAN 80 (Subnet 10.0.0.32/27).
- **Gateway (Virtual IP):** 10.0.0.33 (مُعدّ على Multilayer Switch – Active/Standby HSRP).
- **Broadcast:** 10.0.0.63.
- **Usable IP Addresses:** 30
- **Used IPs:** 14 PCs + 3 reserved addresses (Active Gateway, Standby Gateway, Virtual IP) = 17
- **Available IPs (not used):** 13
- **Access Switch**
 - **Model:** Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports).
 - **Access Ports:**
 - Interfaces Fa0/3 – Fa0/16 configured as Access Mode, assigned to VLAN 80 (for connecting PCs).
 - **Trunk Ports:**
 - Interfaces Fa0/1 – Fa0/2 configured as Trunk Mode, used as uplinks to the Multilayer Switch.
 - **Unused Ports:**
 - Fa0/17 – Fa0/24, Gi0/1 – Gi0/2 → currently unused and available for future expansion.

**** → Floor 3:**

- **Number Of Hosts :** 10 جهاز PC.
- **VLAN:** VLAN 90 (Subnet 10.0.0.112/28).
- **Gateway (Virtual IP):** 10.0.0.113 (مُعدّ على Multilayer Switch – Active/Standby HSRP).
- **Broadcast:** 10.0.0.127.
- **DHCP:** موزع من ال Multilayer Switch (Active).
- **Usable IP Addresses:** 14
- **Used IPs:** 10 PCs + 3 reserved addresses (Active Gateway, Standby Gateway, Virtual IP) = 13
- **Available IPs (not used):** 1
- **Access Switch**
 - **Model:** Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports).
 - **Access Ports:**
 - Interfaces Fa0/3 – Fa0/12 configured as Access Mode, assigned to VLAN 90 (for connecting PCs).
 - **Trunk Ports:**
 - Interfaces Fa0/1 – Fa0/2 configured as Trunk Mode, used as uplinks to the Multilayer Switch.
 - **Unused Ports:**
 - Fa0/13 – Fa0/24, Gi0/1 – Gi0/2 → currently unused and available for future expansion.

→ Multilayer Switch 4 (Active) :

- **Model:** Cisco Catalyst 6500 (Multilayer Switch).

- **Role:** Active switch in HSRP (provides routing between floors of Building 1 and forwards traffic to the Edge Router).
- **VLANs and Interfaces**
 - **VLAN 70 (Floor 1):**
 - Interface IP: 10.0.0.130/28.
 - Virtual IP (HSRP): 10.0.0.129.
 - This virtual IP is the default gateway for all PCs in Floor 1.
 - **VLAN 80 (Floor 2):**
 - Interface IP: 10.0.0.34/27.
 - Virtual IP (HSRP): 10.0.0.33.
 - Default gateway for Floor 2 PCs.
 - **VLAN 90 (Floor 3):**
 - Interface IP: 10.0.0.114/28.
 - Virtual IP (HSRP): 10.0.0.113.
 - Default gateway for Floor 3 PCs.
- **Trunk Interfaces**
 - Gi1/0/1 – Gi1/0/3 configured as Trunk Ports, connecting the Multilayer Switch to the access switches of the three floors.
- **DHCP Configuration**
 - **Pool F7 (Floor 1):**
 - Network: 10.0.0.128/28
 - Default-router: 10.0.0.129
 - DNS: 10.0.0.180
 - Excluded addresses: 10.0.0.129 – 10.0.0.131 (Virtual IP + Active + Standby).
 - **Pool F8 (Floor 2):**
 - Network: 10.0.0.32/27
 - Default-router: 10.0.0.33
 - DNS: 10.0.0.180
 - Excluded addresses: 10.0.0.33 – 10.0.0.35 (Virtual IP + Active + Standby).
 - **Pool F9 (Floor 3):**
 - Network: 10.0.0.112/28
 - Default-router: 10.0.0.113
 - DNS: 10.0.0.180
 - Excluded addresses: 10.0.0.113 – 10.0.0.115 (Virtual IP + Active + Standby).
- **Uplink to Edge Router**
 - Interface Gi1/0/4 is configured as a Routed Port:
 - IP address: 10.0.0.201/30.
 - Connected directly to the Edge Router.
- **HSRP Configuration**
 - **Active Role:** This Multilayer Switch is configured as the Active HSRP device for VLANs 70, 80, and 90.
 - **Standby Role:** The second Multilayer Switch in Building 3 acts as the Standby HSRP device.
 - **Priority:** Set to 150 (higher than the default 100), ensuring that this switch always takes the Active role when available.
 - **Preempt Enabled:** With the preempt command configured, if this Active switch fails and later recovers, it will automatically reclaim the Active role.
- **Routing**
 - **Default Route:** Configured to forward unknown traffic to the Edge Router (Next-Hop: 10.0.0.202).
 - **OSPF:** Enabled for internal routing between VLANs and to share routes with the Edge Router, ensuring connectivity between different buildings.

→ Multilayer Switch 5 (Standby) :

- **Model:** Cisco Catalyst 6500 (Multilayer Switch).
- **Role:** Standby switch in HSRP (provides redundancy for the Active switch, takes over only if the Active fails).
- **VLANs and Interfaces**

- **VLAN 70 (Floor 1):**
 - Interface IP: 10.0.0.131/28.
 - Virtual IP (HSRP): 10.0.0.129.
 - This virtual IP is the default gateway for all PCs in Floor 1.
- **VLAN 80 (Floor 2):**
 - Interface IP: 10.0.0.35/27.
 - Virtual IP (HSRP): 10.0.0.33.
 - Default gateway for Floor 2 PCs.
- **VLAN 90 (Floor 3):**
 - Interface IP: 10.0.0.115/28.
 - Virtual IP (HSRP): 10.0.0.113.
 - Default gateway for Floor 3 PCs.
- **Trunk Interfaces**
 - Gi1/0/1 – Gi1/0/3 configured as Trunk Ports, connecting the Multilayer Switch to the access switches of the three floors.
- **DHCP Configuration**
 - **Pool F7 (Floor 1):**
 - Network: 10.0.0.128/28
 - Default-router: 10.0.0.129
 - DNS: 10.0.0.180
 - Excluded addresses: 10.0.0.129 – 10.0.0.131 (Virtual IP + Active + Standby).
 - **Pool F8 (Floor 2):**
 - Network: 10.0.0.32/27
 - Default-router: 10.0.0.33
 - DNS: 10.0.0.180
 - Excluded addresses: 10.0.0.33 – 10.0.0.35 (Virtual IP + Active + Standby).
 - **Pool F9 (Floor 3):**
 - Network: 10.0.0.112/28
 - Default-router: 10.0.0.113
 - DNS: 10.0.0.180
 - Excluded addresses: 10.0.0.113 – 10.0.0.115 (Virtual IP + Active + Standby).
- **Uplink to Edge Router**
 - Interface Gi1/0/4 is configured as a Routed Port:
 - IP address: 10.0.0.205/30
 - Connected directly to the Edge Router.
- **HSRP Configuration**
 - **Standby Role:** This Multilayer Switch operates as the Standby HSRP device for VLANs 70, 80, and 90.
 - **No Priority/Preempt:** By default, it takes over only if the Active switch fails, and it does not reclaim Active role once the Active comes back online.
- **Routing**
 - **Default Route:** Configured to forward unknown traffic to the Edge Router (Next-Hop: 10.0.0.206).
 - **OSPF:** Enabled to advertise VLAN networks and learn routes from the Edge Router, ensuring inter-building connectivity.

→ Edge Router 2 :

- **Model:** PT-Router (Packet Tracer Default Router)
- **Role:** Connects the internal network of Building 3 to the ISP, applies NAT, and It also acts as the interconnection point between different buildings, ensuring that devices in Building 1, Building 2, and Building 3 can communicate with each other through routing.
- **Interfaces and IPs**
 - FastEthernet 0/0 → 10.0.0.202/30 (connected to Active MLS)
 - FastEthernet 1/0 → 10.0.0.206/30 (connected to Standby MLS)
 - Serial 2/0 → 20.0.0.9/30 (connected to router of ISP)

- Serial 3/0 → 10.0.0.214/30 (connected to router of build2)
- **Routing**
 - **Default Route:** 0.0.0.0/0 → Next-Hop 20.0.0.10 (ISP)
 - **OSPF:** Configured to advertise the routed links to the Multilayer Switches.
This allows all internal subnets of the three floors to reach the router through the MLS devices, and ensures communication between buildings as well as access to the ISP.
↪Advertised networks:
 - 10.0.0.200/30 (link to Active MLS)
 - 10.0.0.204/30 (link to Standby MLS)
 - 10.0.0.212/30 (link for build2)
- **Network Address Translation (NAT)**
 - **Type:** NAT Overload (PAT) :
 - ACL 1 defines the inside subnets permitted for NAT:
 - 10.0.0.128/28 (Floor 1)
 - 10.0.0.32/27 (Floor 2)
 - 10.0.0.112/28 (Floor 3)
 - **Configuration:**
 - NAT Inside Interfaces: FastEthernet 0/0, FastEthernet 1/0
 - NAT Outside Interface: se2/0
 - Translation: **All inside networks share the public IP of Gi9/0 (20.0.0.9) using overload.**

```
access-list 1 permit 10.0.0.128 0.0.0.15 -
access-list 1 permit 10.0.0.32 0.0.0.31 -
access-list 1 permit 10.0.0.112 0.0.0.15 -
ip nat inside source list 1 interface se2/0 overload -
int se2/0 -
ip nat outside -
int f0/0 -
ip nat inside -
int f1/0 -
ip nat inside -
```

◆ BUILD ISP

- **Role:** This building represents the **Internet Service Provider (ISP)**. It provides external services (Web and FTP) that can be accessed from the enterprise network (Buildings 1, 2, and 2).
- **Network:** 192.168.1.0/24
- **Default Gateway:** 192.168.1.1 (configured on PT-Router).

➔ Devices:

- **Server 1 (Web Server):**
 - **IP Address:** 192.168.1.2/24
 - **Service:** Web Browser (HTTP)
 - Accessible by clients from the enterprise network.
- **Server 2 (FTP Server):**
 - IP Address: **192.168.1.3/24**
 - Service: FTP
 - Authentication required for access:
 - Username: said
 - Password: `12345
- **Access Switch**

- **Model:** Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports).
- **Access Ports:**
 - Interfaces Fa0/1 – Fa0/3 configured as Access Mode by default.
- **Unused Ports:**
 -
 - Fa0/4 – Fa0/24, Gi0/1 – Gi0/2 → currently unused and available for future expansion.
- → **ISP Router:**
- **Model:** PT-Router (Packet Tracer Default Router)
- **Interfaces and IPs**
 - FastEthernet 0/0 → 20.0.0.2/30 (connected edge rw of build1)
 - FastEthernet 1/0 → 20.0.0.6/30 (connected to edge rw of build2)
 - Serial 2/0 → 20.0.0.10/30 (connected to edge rw of build3)
 - g9/0 → 192.168.1.0/24 (connected to ISP External Services Network)
- **Network Address Translation (NAT)**
 - **Type:** PAT (Port Address Translation – one-to-one mapping per building)
 - **ACL 1** defines the inside network permitted for NAT:
 - 192.168.1.0/24 (ISP External Services Network – Web & FTP Servers)
 - **Configuration:**
 - **NAT Inside Interface:** GigabitEthernet 9/0.
 - **NAT Outside Interfaces:** FastEthernet 1/0, Serial 2/0 and FastEthernet 0/0.
 - **Translation:**
 - **Each building is mapped to a dedicated public IP when accessing the ISP:**
 - 20.0.0.2 → Building 1
 - 20.0.0.6 → Building 2
 - 20.0.0.10 → Building 3
 - ⚠ This ensures that traffic from each building uses its own unique public IP.
 - ⚠ In case of a link failure, a building's traffic can be rerouted via another available public IP.
 - **Static NAT (Port Forwarding):**
 - **Web Server (192.168.1.2:80) accessible via:**
 - 20.0.0.2:80, 20.0.0.6:80, 20.0.0.10:80
 - **FTP Server (192.168.1.3:20–21) accessible via:**
 - 20.0.0.2:20–21, 20.0.0.6:20–21, 20.0.0.10:20–21

```

access-list 1 permit 192.168.1.0 0.0.0.255 -
ip nat inside source list 1 interface f0/0 overload -
ip nat inside source static tcp 192.168.1.2 80 20.0.0.2 80 -
ip nat inside source static tcp 192.168.1.3 20 20.0.0.2 20 -
ip nat inside source static tcp 192.168.1.3 21 20.0.0.2 21 -
int g9/0 -
ip nat inside -
int f1/0 -
ip nat outside -
ip nat inside source list 1 interface f1/0 overload -
ip nat inside source static tcp 192.168.1.2 80 20.0.0.6 80 -
ip nat inside source static tcp 192.168.1.3 20 20.0.0.6 20 -
ip nat inside source static tcp 192.168.1.3 21 20.0.0.6 21 -
int f1/0 -
ip nat outside -
ip nat inside source list 1 interface se2/0 overload -
ip nat inside source static tcp 192.168.1.2 80 20.0.0.10 80 -
ip nat inside source static tcp 192.168.1.3 20 20.0.0.10 20 -
ip nat inside source static tcp 192.168.1.3 21 20.0.0.10 21 -

```

◆ ### Verification / Testing

- **Port Security**
 - On interface Fa0/5 , if another device is connected (e.g., replacing PC1), the port will automatically shut down and block the new device.
- **Access Control List (ACL)**
 - PC0 (Static IP 10.0.0.4) is not able to access the **FTP Server** due to the applied ACL.
 - However, PC0 can still access other services (Web, Email, DNS) without restriction.
- **HSRP Active/Standby Verification (Telnet)**
 - By using **Telnet** to connect to the **Virtual IP (10.0.0.1)** from any PC in Building 1, the session will always be redirected to the **Active Multilayer Switch**.
 - This allows quick verification of which switch is currently **Active** and which one is **Standby**.
- **DNS Service**
 - From any PC in the network, the DNS can resolve the domain www.mahmoud.com to the correct server IP.
- **Web Service**
 - Using a web browser on any PC, access the Web Server via its IP: **10.0.0.180**.
- **FTP Service**
 - From any PC in the topology, FTP access is possible using the server IP **10.0.0.180**.
 - For verification, a file named **net.txt** was uploaded to the FTP server from **PC49** and successfully downloaded by **PC56**.
- **Email Service**
 - Email functionality tested between **PC48** and **PC55** (send/receive).
- **Inter-Building Connectivity**
 - From any PC in Building 1, a successful **ping** was made to PCs in other buildings, confirming OSPF routing and connectivity.
- **NAT Verification**
 - To test NAT, from any building try accessing an **unknown external network**.
 - Each building goes out to the internet using its **dedicated Public IP**:
 - Building 1 → **20.0.0.2**
 - Building 2 → **20.0.0.6**
 - Building 3 → **20.0.0.10**
 - Using a web browser or FTP client, external ISP services can be accessed via these public IPs.

◆ Conclusion

- This project successfully demonstrates the design and implementation of a multi-building enterprise network.
- **Key highlights include:**
 - VLAN segmentation and DHCP services per floor.
 - Redundancy and load balancing with HSRP on multilayer switches.
 - Secure access using Port Security and ACLs.
 - Full inter-building connectivity via OSPF.
 - External connectivity to the ISP through NAT (PAT) and dedicated public IPs.
 - Hosting and accessing services (Web, DNS, FTP, Email) on ISP servers.
- Through step-by-step testing, all network functionalities were verified, ensuring reliability, redundancy, and scalability of the design.

Topology Screenshot:

